Rajshahi University of Engineering & Technology
Department of Electrical & Elctronic Engineering

Design and Implementation of an Intelligent Autonomous Railway Crossing Safety Framework with Multi-Sensor Redundancy

ABSTRACT

This project presents an intelligent, cost-effective autonomous railway level crossing safety system using Arduino Uno. The system integrates multi-sensor redundancy — primary HC-SR04 ultrasonic sensors and backup IR barrier sensors — with a state-based control logic engine. Upon train detection, servo motors actuate physical barriers automatically. A Bluetooth telemetry module streams real-time tracking data and fault alerts to a companion Android application. The design eliminates human error dependency, provides built-in fail-safe switching, and offers a scalable solution for regional railway infrastructure modernization within a minimal budget of BDT 2,910.

PROJECT BACKGROUND

Conventional level crossing systems in developing regions rely on manual operation or single-variable sensors, making them vulnerable to:

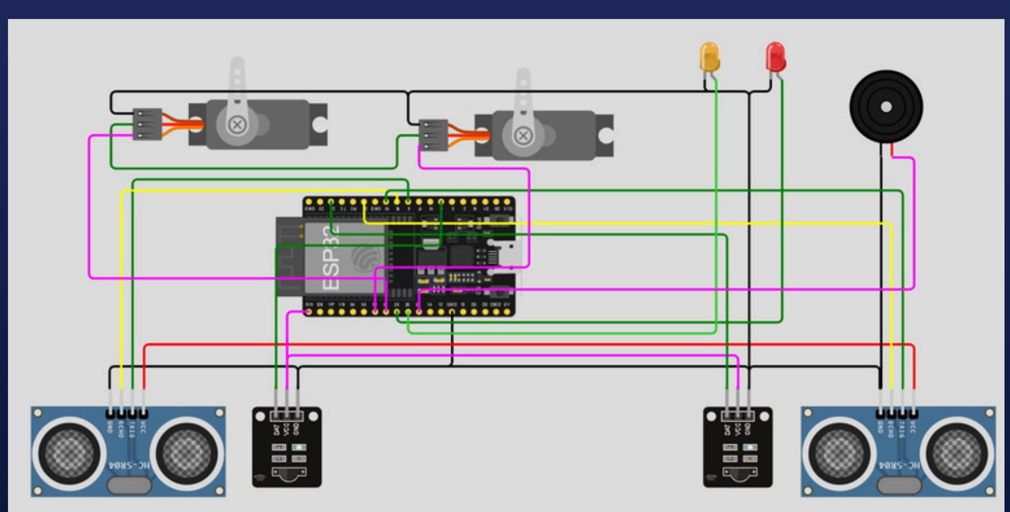
- Human error and mechanical latency
- Single-point sensor failure (dust/debris blinding)
- Environmental interference degrading reliability
- Prohibitive cost of radar/computer-vision systems

This project bridges the safety gap by delivering a redundant, autonomous, and fault-tolerant level crossing system at a fraction of the cost of commercial alternatives.

OBJECTIVES

- Design and develop an autonomous railway level crossing system
- Implement reliable train detection via ultrasonic and IR sensors
- Develop safe, deterministic gate control logic
- Establish wireless remote monitoring and alert system
- Enhance crossing safety while reducing human dependency

Circuit Simulation



Methodology

The system follows a systematic execution plan spanning hardware integration, logic programming, and performance validation. Development proceeds through five key phases:

Phase 1 — Component Selection & Integration
↓
Phase 2 — Hardware Interfacing (I/O Mapping)
↓
Phase 3 — Software Logic & Programming
↓
Phase 4 — App Integration & Interface Design
↓
Phase 5 — Testing & Calibration

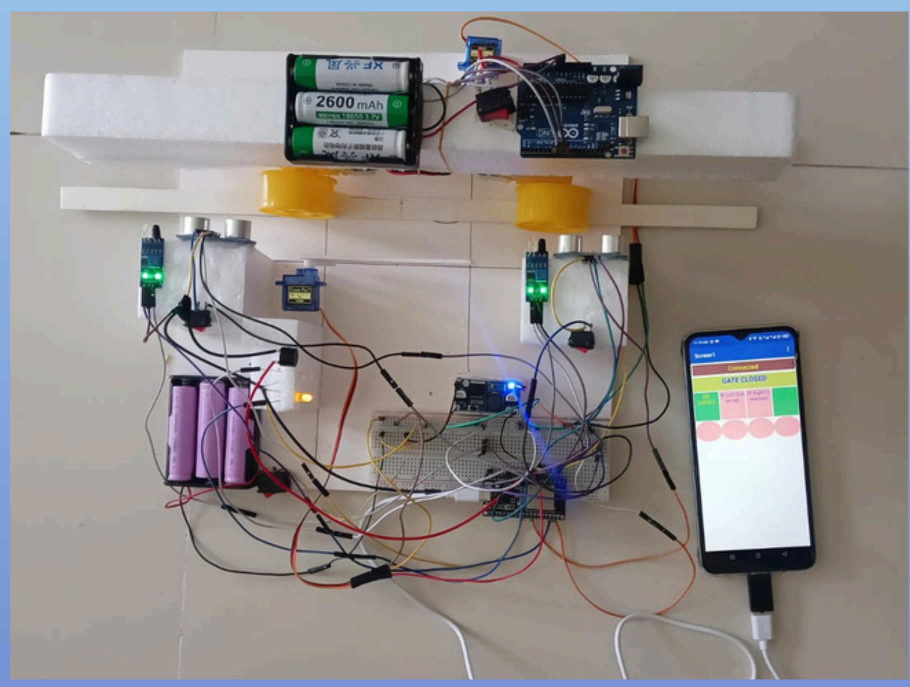
REQUIRED COMPONENTS

Component	Spec/Model	Qty
ESP-32	R3+USB Cable	1
Ultrasonic Sensor	HC-SRO4	2
IR Sensor Module	Obstacle Avoidance	3
Servo Motor	SG-90	2
Bluetooth Module	HC-05	1
12C LCD Display	16x2	1
Piezo Buzzer	5V	1
LEDs + Resistors	220 ohm R/G	1 set
Breadboard + Wires	MB102	1 set
BAttery/Adapter	5V , 2A	1
	Total Budget:	BDT 2910

Conclusion

The system successfully demonstrates that a cost-effective, fault-tolerant, and autonomous railway level crossing solution is achievable using readily available embedded hardware — delivering deterministic safety without dependence on high-cost infrastructure.

Circuit Setup



Result and Discussion

Deterministic Train Detection

- High-accuracy acoustic time-of-flight profiling provides real-time directional tracking of approaching rolling stock via the dual HC-SR04 array.

Rapid-Response Barrier Modulation

- Immediate servo motor gate closure within a calculated time threshold prevents pedestrian or vehicle track breach before train arrival.

Fault-Tolerant Redundancy

- Instantaneous hand-off to backup IR barrier sensors activates when primary ultrasonic sensors suffer environmental blinding or hardware failure.

Proactive Telemetry & Diagnostics

- Real-time extraction of track occupancy status, kinematic metrics, and cross-sensor discrepancy alerts delivered to the Android application layer.

Future Scopes

- Replace Bluetooth with GSM/4G for long-range cloud telemetry and remote control over cellular networks.
- Integrate computer vision (Raspberry Pi + camera) for vehicle density estimation and pedestrian detection at crossings.
- Develop a central IoT dashboard for multi-crossing fleet management across a regional railway network.
- Add ML-based predictive maintenance to forecast sensor degradation before physical failure occurs.
- Extend the system for two-track crossings with simultaneous multi-directional train monitoring capability.